Alina Shalukhina

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Ciência ID: 4C15-B301-262C

Identification

Citizenship: *Ukraine*
Date of birth: October 20, 1996
Position: 4th-year PhD student at *NOVA University Lisbon*, Portugal
Research interests: Harmonic Analysis, Functional Analysis, Operator Theory

Education

PhD **NOVA University Lisbon, NOVA FCT** *Portugal*
Major in Analysis and Geometry 2022 – present
Thesis: *The maximal operator on variable Lebesgue spaces over spaces of homogeneous type*
Advisor: Oleksiy Karlovych

MSc **Odesa I. I. Mechnikov National University** *Ukraine*
Major in Mathematical Analysis 2017 – 2018
Thesis: *Reverse inequalities*
Advisor: Anatolii Korenovskyi
Grade: 98.3/100 with honors

BSc **Odesa I. I. Mechnikov National University** *Ukraine*
Major in Mathematical Analysis and Mathematics Teaching 2013 – 2017
Grade: 97/100 with honors

Publications

Journal articles

1. T. Kopaliani, A. Shalukhina, and G. Tephnadze. "Maximal operator on $L^{p(\cdot)}$ spaces over spaces of homogeneous type: Local boundedness," *Georgian Math. J.* (2026), Published online. DOI: 10.1515/gmj-2026-3048.

2. A. Shalukhina. "The maximal function on spaces of homogeneous type, or adjacent dyadic cubes do good," *Anal. Math. Phys.* 16.3 (2026), Paper no. 63. DOI: 10.1007/s13324-026-01215-5.

3. A. Shalukhina. "Self-improving boundedness of the maximal operator on quasi-Banach lattices over spaces of homogeneous type," *J. Math. Anal. Appl.* 548.2 (2025), Paper no. 129419. DOI: 10.1016/j.jmaa.2025.129419.

4. O. Karlovych and A. Shalukhina. "A necessary condition for the boundedness of the maximal operator on $L^{p(\cdot)}$ over reverse doubling spaces of homogeneous type," *Anal. Math.* 51.1 (2025), 241–248. DOI: 10.1007/s10476-024-00053-6.

5. A. Shalukhina. "On the extension of the reverse Hölder inequality for power functions on the real axis," *Real Anal. Exch.* 47.2 (2022), 425–434. DOI: 10.14321/realanalexch.47.2.1653382423.

Book chapters

1. A. Shalukhina. "On Interpolation of Variable Lebesgue Spaces over Spaces of Homogeneous Type," to appear in *Tbilisi Analysis and PDE Seminar* (Trends in Mathematics 20), Birkhäuser, Cham, 2026. URL: link.springer.com/book/9783032218063.

2. O. Karlovych and A. Shalukhina. "Maximal Noncompactness of Singular Integral Operators on L^2 Spaces with Some Khvedelidze Weights," in *Operator and Matrix Theory, Function Spaces, and Applications* (Operator Theory: Advances and Applications 295), Birkhäuser, Cham, 2024, pp. 279–295. DOI: 10.1007/978-3-031-50613-0_12.

3. O. Karlovych and A. Shalukhina. "Maximal Noncompactness of Singular Integral Operators on L^p Spaces with Power Weights," in *Tbilisi Analysis and PDE Seminar* (Trends in Mathematics 7), Birkhäuser, Cham, 2024, pp. 87–97. DOI: 10.1007/978-3-031-62894-8_9.

Research Grant

PhD Scholarship UI/BD/154284/2022 — *Portugal*
Funder: *FCT – Fundação para a Ciência e a Tecnologia* — 2022 – present
Host: Center for Mathematics and Applications – NOVA Math

Distinction

Member of the EMS Young Academy (EMYA) — *Europe*
European Mathematical Society — 2027 – 2030
Selected for the 2027–2030 cohort

Conferences and Workshops

2025 **47th Summer Symposium in Real Analysis**
Universidad Complutense de Madrid, *Spain*
Talk: *The Hardy–Littlewood maximal function on spaces of homogeneous type*

12th International Conference on Harmonic Analysis and Partial Differential Equations
El Escorial, Spain
Poster: *The Hardy–Littlewood maximal function on spaces of homogeneous type*

XV International Conference of the Georgian Mathematical Union
Batumi Shota Rustaveli State University, *Georgia*
Talk: *The maximal operator on variable $L^{p(\cdot)}$ spaces over spaces of homogeneous type*

Tbilisi Analysis & PDE Workshop
University of Georgia, *Tbilisi, Georgia*
Talk: *Interpolation of variable Lebesgue spaces over spaces of homogeneous type*

Lisbon Young Mathematicians Conference
ISEG – Lisbon School of Economics and Management, *Portugal*
Talk: *Maximal operator on variable Lebesgue spaces: Euclidean vs. Spaces of homogeneous type*

2024 **Young Researchers in Functional Analysis and Mathematical Physics'24**
Universidade da Madeira, *Funchal, Portugal*
Talk: *Self-improving property of the Hardy–Littlewood maximal operator over spaces of homogeneous type*

Tbilisi Analysis & PDE Workshop
University of Georgia, *Tbilisi, Georgia*
Talk: *Self-improving property of the Hardy–Littlewood maximal operator over spaces of homogeneous type*

Lisbon Young Mathematicians Conference
Instituto Superior Técnico, *Portugal*
Talk: *Self-improving boundedness of the Hardy–Littlewood maximal operator over spaces of homogeneous type*

2023 **3rd Women in Mathematics Meeting**
Universidade de Coimbra, *Portugal*
Talk: *On the extension of the reverse Hölder inequality for power functions on the real axis*

Lisbon Young Mathematicians Conference
Universidade Aberta, *Portugal*
Talk: *On the extension of the reverse Hölder inequality for power functions on the real axis*

Invited Seminars

University of Nyíregyháza *Hungary*
Seminar at the Institute of Mathematics and Computer Sciences 2026
Talk: *The maximal function on spaces of homogeneous type, or adjacent dyadic cubes do good*

Kutaisi International University *Georgia*
Seminar at the Giorgi Nikoladze Scientific Research Center 2026
Talk: *The maximal function on spaces of homogeneous type, or adjacent dyadic cubes do good*

NOVA Math, NOVA University Lisbon *Portugal*
Seminar of Analysis 2023 – 2026

- **2026** | Talk: *The maximal function on spaces of homogeneous type, or adjacent dyadic cubes do good*
- **2025** | Talk: *On interpolation of variable Lebesgue spaces over spaces of homogeneous type*
- **2024** | Talk: *How the maximal operator's self-improving property translates into spaces of homogeneous type*
- **2023** | Talk: *The maximal operator: when is it bounded on variable Lebesgue spaces?*

Czech Technical University in Prague *Czech Republic*
MAFIA – Methods of Algebra and Functional analysis In Applications 2024
Talk: *The maximal operator, the variable $L^{p(\cdot)}$, and the bizarre geometry of spaces of homogeneous type*

CIDMA, Universidade de Aveiro *Portugal*
Functional Analysis and Applications Seminar 2024
Talk: *A necessary condition for the boundedness of the maximal operator on $L^{p(\cdot)}$ over reverse doubling spaces of homogeneous type*

Organization of Events

XVI International Conference of the Georgian Mathematical Union *Georgia*
Member of the Organizing Committee 2026
Host: Batumi Shota Rustaveli State University

- *Allocated abstracts to thematic sessions and compiled participant statistics*
- *Managed international registration and prepared welcome kits*

Teaching & Analytic Experience

Online Mathematics Tutor *Ukraine*
Freelance teaching and consulting 2020 – 2022
Coached 18 high school students in algebra, geometry & elements of calculus

Mathematics Tutor in Odesa *Ukraine*
Full-time in-person teaching (self-employed) 2019 – 2020

- *Taught algebra and geometry to individual secondary school students*
- *Achieved a 100% success rate in preparing 12 students for national university entrance exams*

Assistant Analyst, Marine Biology Research Group *Ukraine*
Odesa I. I. Mechnikov National University 2018

- *Conducted statistical Fourier-descriptors-based analysis of fish otolith shapes to distinguish round goby populations in the Black Sea*
- *Processed digital images in Shape v. 1.3, extracting harmonic contours for statistical analysis*

Associated work: V. Zamorov, Y. Leonchyk, M. Zamorova, N. Matviienko & A. Shalukhina. "Estimation Method of Round Goby Natural Mortality Rates Depending on Fish Length," *Proceedings of FABA 2018*, Ankara, Turkey, p. 105.

Languages

Ukrainian, Russian: Native
English: Advanced
Portuguese: Elementary